

# URLLC and eMBB in 5G Industrial IoT: A Survey

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**Abstract**—Fifth generation (5G)-industrial Internet of things (IIoT) is the integration of IIoT and a private 5G network. The IIoT is a concept that involves incorporating smart objects, gadgets, and solutions into cutting-edge industrial operations to increase reliability, efficiency, and over-production costs. Furthermore, the integration of IIoT and 5G/beyond 5G (B5G) provides the potential for ubiquitous and instantaneous connectivity. The 5G architecture can handle the IIoT's stringent ultra-low latency, real-time processing, high data-rate, nearby storage, and reliability requirements. A new era of economic growth is predicted for IIoT assisted 5G/B5G wireless networks. It should be noted that the majority of the work in IIoT is focused on the architecture, with reliability and throughput being largely ignored. This paper provides a comprehensive review of B5G assisted IIoT wireless networks, with a focus on enhanced mobile broadband (eMBB) and ultra-reliable low latency communication (URLLC) services. Furthermore, it provides insights into various applications and key enabling technologies from the perspective of URLLC, eMBB and their tradeoff.

**Index Terms**—fifth generation (5G), beyond 5G (B5G), sixth generation (6G), industrial Internet of things (IIoT), ultra-reliable low latency communication (URLLC), enhanced mobile broadband (eMBB).

## I. INTRODUCTION

THE emergence of industrial Internet of things (IIoT) has revolutionized industrial operations such as manufacturing and production by automating a huge number of connected components and devices. IIoT is a sub-category of Internet of things (IoT) that focuses on the use of IoT techniques and technologies [1]–[4] in industries such as smart cities, smart transportation, smart grid, smart health services, forestry, food, weather, agriculture, monitoring, and surveillance [5]–[18]. The machines/devices in IIoT are connected to capture smartness and autonomy in legacy systems [19]–[21]. It is estimated that by the end of 2030, roughly 80 billion devices will be connected to the Internet [22]–[24]. In IIoT, an increased degree of connectivity is involved, which has special requirements for high reliability, low latency, high speed, flexible, and secure communication [25]–[27].

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Forth generation (4G) wireless networks has gained popularity over the past decade as a reliable communications technology [27]–[29]. Many industrial applications are supported by long term evolution (LTE) and wireless fidelity (WiFi), but with more demanding performance needs from users in terms of reliability, latency, and throughput, fifth generation (5G) has emerged as a promising alternative that can satisfy these requirements [30]. 5G started becoming commercially available in 2019. It is used a wide range of applications from IoT to smart home to Industry(4.0). Moreover, the International Telecommunication Union (ITU) defines three usage scenarios: ultra-reliable low latency communication (URLLC), enhanced mobile broadband (eMBB), and massive machine type communication (MTC) (mMTC) [31]–[34]. To sustain/achieve the competitive requirements for the applications of wireless networks, the industry and academia have started to conceptualize the next generation of wireless communication systems (sixth generation (6G)) [35]–[37]. The 6G is expected to have a data rate of 20 Gbps, admit  $10^6$  devices per  $\text{km}^2$ , and have a latency/delay of less than 1 ms. Thus, 5G/6G enhancements are capable of providing the essential services and meeting the performance metrics of IIoT.

For integration of 5G/6G with IIoT, a private 5G network can be designed and deployed [38]–[40]. The requirement for a private network has been addressed in the 5G standards, rather than being an add-on capability in previous generations. The primary motive for establishing a private 5G network is the guaranteed coverage as well as the better performance profile over legacy wireless technologies. Because most industries are located in distant places where public network coverage is limited or non-existent, a private network can provide the assured service required by IIoT. It is a secure network customized for a certain industry, with a dedicated radio access network (RAN) and core mobile network for mobile communication. As a result, a private 5G network for IIoT known as 5G-IIoT could be a promising solution.

5G-IIoT can be used to complement the existing communication solutions for IIoT and can be used based on application requirements. The two most characteristic features of 5G-IIoT are a) URLLC, which requires a latency of less than 1 ms and a reliability of 99.99%, and b) eMBB, which requires a data rate of giga bits per second. Therefore, 5G-IIoT may achieve extremely high data-rates, low latency with wide coverage, and relatively low power consumption [41], [42].

## A. Existing Surveys

Table I provides a summary of the existing surveys on the IIoT. Xu *et al.* [43] outlines a comprehensive survey work for IIoT. This work covers current research in IoT technology,

TABLE I: An overview of existing surveys on IIoT. Rel.: Reliability, Tp.: Throughput, Appl.: Applications, Chall.: Challenges.

| Ref.             | Title                                                                                   | 5G / 6G | Techniques | Rel. (URLLC) | Tp. (eMBB) | Trade-off | Appl. | Chall. |
|------------------|-----------------------------------------------------------------------------------------|---------|------------|--------------|------------|-----------|-------|--------|
| [43]             | IIoT in industries: A survey                                                            |         | ✓          |              |            |           | ✓     | ✓      |
| [44]             | IIoT: Challenges, opportunities, and directions                                         |         |            |              |            |           |       | ✓      |
| [19]             | Industrial IIoT in 5G environment towards smart manufacturing                           | ✓       | ✓          |              |            |           |       | ✓      |
| [45]             | Communication protocols of an IIoT environment: A comparative study                     |         | ✓          |              |            |           |       |        |
| [46]             | Creating values out of IIoT: An industrial perspective                                  |         |            |              |            |           |       | ✓      |
| [47]             | 5G URLLC implementation challenges and operational issues with IIoT devices             | ✓       |            | ✓            | ✓          |           | ✓     |        |
| [20]             | URLLC wireless communication: Tail, risk and scale                                      |         |            | ✓            |            | ✓         | ✓     |        |
| [21]             | A comprehensive survey on IIoT towards 5G wireless systems                              | ✓       | ✓          |              |            |           | ✓     | ✓      |
| [48]             | A Energy-Efficient IIoT: Overview and Open Issues                                       |         | ✓          |              |            |           |       | ✓      |
| <b>This work</b> | URLLC and eMBB Support for IIoT: Techniques, Challenges, Applications and Research Gaps | ✓       | ✓          | ✓            | ✓          | ✓         | ✓     | ✓      |

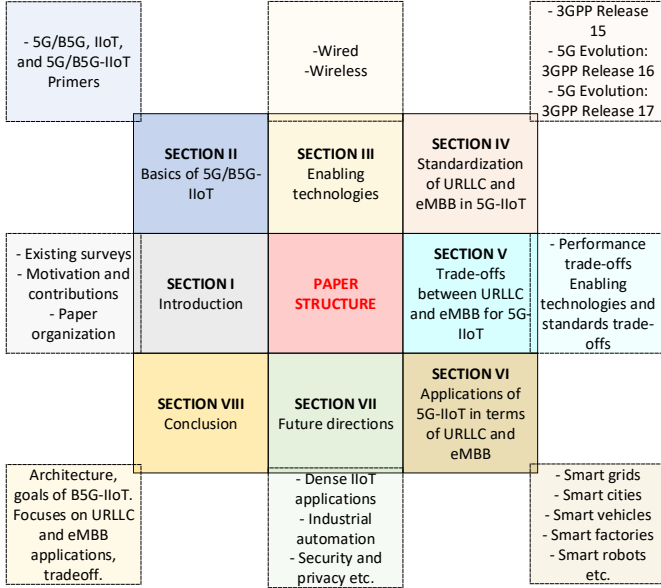


Fig. 1: Structure of paper.

applications, and industry issues. In [44], a brief introduction to IIoT, IIoT, and industry 4.0, with emphasis on performance, parameters, research efforts, and future directions in the domain is provided. The implementation methods, manufacturing technologies, and scenarios of beyond 5G (B5G)-IIoT based cyber-physical manufacturing systems (CPMS) are discussed in detail in [19]. Event based polling protocols for IIoT are briefly discussed in [45]. Various applications in the domains of IIoT cloud platform with industrial perspective are debated in [46]. Further, the challenges of cloud based IIoT are inves-

tigated to generate potentially more business opportunities in industries. In [47], the progress in third generation partnership project (3GPP), and implementation issues of URLLC in IIoT devices is briefly addressed. In [20], the main focus is on requirements, techniques and methodologies related to URLLC to ensure the goals of 5G wireless networks. A comprehensive survey of IIoT in 5G system is provided in [21]. It provides a review of enabling and emerging technologies of 5G in IIoT with research gaps. Finally, in [48], challenges and technologies in perspective of energy efficiency are discussed for IIoT.

### B. Motivation and Contributions

Although there are current surveys on IIoT, based on extensive literature search and to the best of the authors knowledge, none of them focus on the IIoT reliability and throughput aspects. URLLC and IIoT have a lot of potential and a promising future in 5G-IIoT. However there is a trade-off between both technologies, which motivated us to investigate IIoT features and requirements, as well as its enabling techniques. We also cover the trade-offs and challenges encountered by IIoT in URLLC and eMBB, as well as future directions.

The following summarizes the major contributions of this work:

- Presents a comprehensive literature review for IIoT in 5G to determine how 5G fits the requirements of IIoT. The prime objective of this work is to identify research gaps in the URLLC and eMBB aspects of IIoT in 5G.
- Provides insights into diverse applications and requirements, where we investigate the key enabling technologies for 5G-IIoT, particularly the communication technologies for URLLC and eMBB.

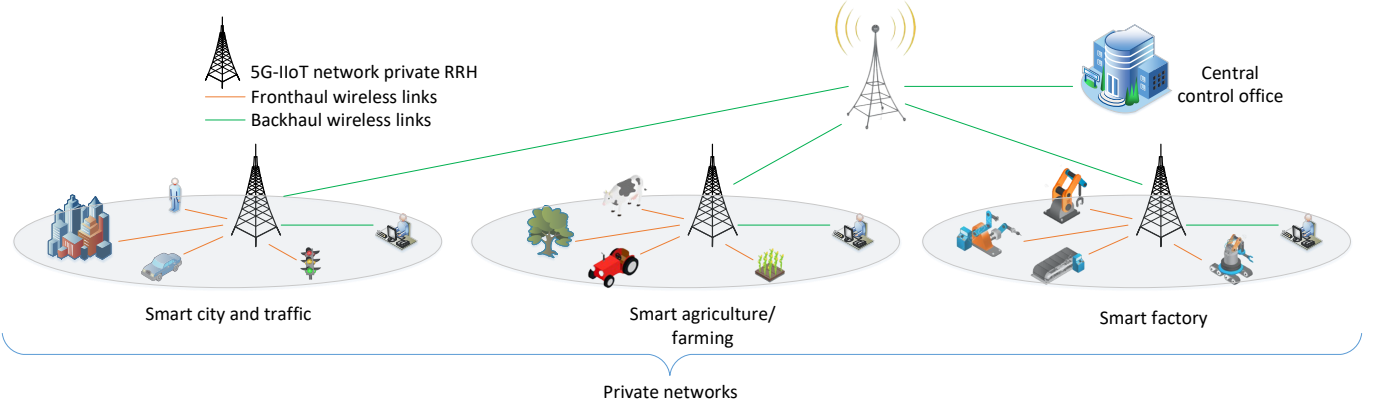


Fig. 2: 5G-IIoT private network.

- Examines the trade-off between URLLC and eMBB.
- Outlines the challenges and future directions for 5G-IIoT for URLLC and eMBB.

### C. Paper Organization

The organization of the paper is as follows: The primer of B5G and IIoT and their relationship is briefly discussed in section II. The enabling techniques for URLLC and eMBB, and their role in IIoT are provided in Section III. Section IV presents 5G standardization for IIoT with respect to URLLC and eMBB. Section V describes the trade-off between URLLC and eMBB. The applications of B5G-IIoT are discussed in Section VI. The challenges and future directions are debated in Section VII. The paper is summarized in Section VIII. The overall structure of the paper is shown in Fig. 1.

## II. BASICS OF 5G/B5G-IIoT

The core architecture of 5G/B5G and IIoT is covered in this section, as well as the 5G-IIoT design and objectives.

### A. 5G/B5G Primer

The ecosystem of 5G/B5G connects the entire society through digital devices and gadgets, driving the economic development of innovative services. Aside from network virtualization, the flexible and dynamic architecture of 5G/B5G offers numerous access technologies with unlicensed and licensed users and devices. The 5G/B5G network may be scaled and extended in cost-effective and automated models to support revolutions in the automotive, manufacturing, utilities, transportation, public safety, healthcare, media, and other industries. The 5G/B5G architecture supports a wide range of advanced access technologies, including WiFi, B5G, new radio (NR), low power wide area (LPWA), and cellular with inter-access mobility [26], [49], [50]. The main features are built on software-defined-networking (SDN) and cloud/virtualization native principles, with full orchestration to provide flexible deployments.

Potential use cases for B5G with varying requirements have been identified by standard bodies and industrial parties. These prospective use cases can be divided into three categories:

- **eMBB:** can be used to service large-scale events and compact metropolitan areas that have high data rate requirements but restricted bandwidth. Virtual and augmented reality, smart offices, 8K/4K video streaming, and cloud applications all require broadband connectivity with a minimum data rate of 50 Mbps everywhere. Furthermore, mobile enabled eMBB services are necessary for enhanced navigation, in-vehicle infotainment, telematics support for diagnostics and safety, and commercial aircraft on-board entertainment.
- **URLLC:** Mission-critical communication is required for emergency services such as disaster response, public safety, and location services. URLLC provides real-time services for scenarios requiring a response time of less than 1 ms. Among the use cases covered by URLLC are robotic control-based industrial automation, autonomous driving, remote surgery, and drones [51].
- **Massive IoT (mIoT):** Long-range, low-cost, and ultra-energy-sensitive devices that require infrequent and periodic connectivity from cloud applications and remote locations are examples of mIoT. LPWA cellular technologies were provided in 3GPP release-13 with long term evolution for machine (LTE-M) and narrowband Internet of things (NB-IoT), and further advances were proposed in release-14, which are aligned with the improvements of the 5G/B5G architecture.

### B. IIoT Primer

The emergence of digital and smart manufacturing in industries aims to merge operational technology (OT) and information technology (IT). In general, the IIoT will connect industrial assets, such as machinery and control systems, to the IT and business processes. This integration results in massive data generation and collection. This data could be used to develop analytic solutions to improve industrial operations [52]. Smart manufacturing, on the other hand, concentrates primarily on the production stage of a smart product's life-cycle, with the aim of responding quickly and dynamically to demand changes. As a result, IIoT has an impact on the industrial value chain and meets the standards for smart manufacturing. In general, the machines in IIoT are meant

to communicate autonomously with one another. The IIoT comprises situations such as legacy monitoring applications, i.e., process monitoring in manufacturing plants, and novel ways for self-organizing systems, i.e., autonomic industrial plants with minimal human intervention.

Typically, IIoT systems are envisioned as a layered modular architecture of digital technologies. The physical components, such as cyber-physical systems (CPS), sensors, or machines, are referred to as the device layer. The network layer consists of physical network buses, cloud computing, and communication protocols that collect and transfer data to the service layer, which consists of applications that transform and combine data to display information on the driver dashboard. The content layer, also known as the user interface, is the topmost layer of the stack. The user interface devices, such as computer screens, point of sale (PoS) stations, tablets, smart glasses, and smart surfaces, are included in this layer. The service layer follows, which includes applications and software for analyzing collected data and transforming it into usable information. It then proceeds to the network layer, where several communication protocols such as WiFi, Bluetooth, and low range (LoRa) operate. The device layer contains the physical hardware of the IIoT network, including devices such as CPS, machines, and sensors.

Standard bodies and industry stakeholders have identified a variety of potential use cases for IIoT. The following are the general requirements and objectives of IIoT:

- Quality of service (QoS) requirement with low latency, and ultra high reliability is vital.
- Low cost scalable network with esteemed security and privacy is desired.
- The emerging standards should be smoothly implemented and integrated on the IIoT devices in a flexible way.
- Inter- and Intra-connection of networks and IoT devices should be possible frequently.

### C. Does 5G/B5G Satisfy IIoT Requirements?

After understanding the fundamental and necessary needs of IIoT, it is critical to establish if present and future wireless communication standards meet these requirements. The current standard, 5G, provides use cases that make industrial systems more flexible, beneficial, and autonomous while meeting QoS requirements of both 5G/B5G and IIoT. Some of the goals that can be reached with B5G in IIoT are as follows:

- Optimized services can be provided through the use of a dedicated network and unified connectivity.
- The standard guarantees 99.999% reliability with less than 1 ms latency in radio frequency (RF) environments.
- It substitutes fixed wired Ethernet lines with re-configurable communication technology.
- Private networks can operate on both licensed and unlicensed spectrum.

### D. 5G/B5G-IIoT Primer

The system model in Fig. 2 depicts the high layer connectivity found in a 5G-IIoT network. 5G/B5G-IIoT is an integration

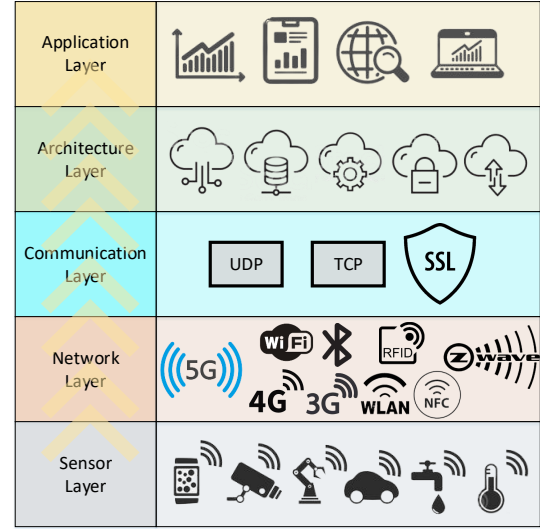


Fig. 3: 5G-IIoT architecture.

of 5G/B5G with IIoT over a private 5G network. The inclusion of 5G/B5G in IIoT can provide the required ubiquitous and instantaneous connectivity for IIoT applications. A private network is one that has its own dedicated RAN and core mobile network. It requires a framework to process and analyze the obtained data using inter- and intra-industrial/device units. As shown in Fig. 3, the B5G-IIoT architecture is primarily separated into five levels [26], [27], [50], [53].

- **Application Layer** This layer includes IIoT applications in B5G, such as smart grid, smart factory, smart city, smart cars, traffic management system, health, and education. Devices and machines that send data via the internet are also included in the application layer.
- **Architecture Layer** This layer incorporates big data analytic (BDA) cloud computing and edge computing for data processing.
- **Communication Layer** The communication layer of the IIoT network transports information between all layers and is considered a critical component of the network.
- **Network Layer** This layer includes communication technologies such as low power wide area network (LPWAN), Bluetooth, WiFi, and 802.11.x, depending on the range and various required factors.
- **Sensor Layer** In the physical layer, sensors and actuators are available. These sensors are in charge of collecting data and transmitting it to the upper layer, i.e., the network layer.

The modern industrial industry is undergoing a technological transformation. Meanwhile, the IIoT proposes new communication technology requirements. Every IIoT application has essential design goals to improve QoS requirements. The key design goals for these critical IIoT applications are depicted in Fig. 4. The key design objectives are briefly stated here.

- **Energy:** Typically, IoT devices are classified as low power devices (LPDs), with limited on-board energy sources. As a result, in order to improve the performance of the IIoT network, these LPDs should make efficient use

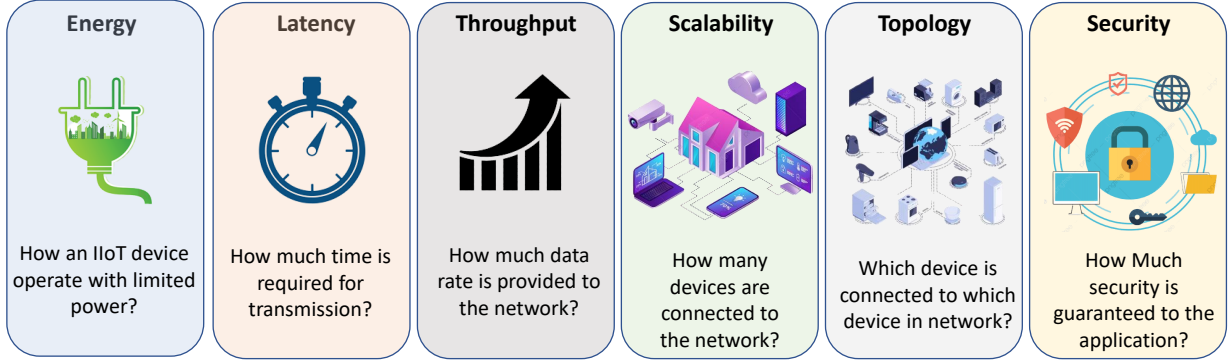


Fig. 4: Design goals for 5G-IIoT.

of their available power resources to extend their life.

- **Latency:** In general, the B5G-IIoT applications are time-sensitive. As a result, a minimum end-to-end latency of less than 1 ms must be reached in all sorts of processing, computation, and propagation in the B5G-IIoT applications.
- **Throughput and coverage:** The QoS requirements vary for different B5G-IIoT applications and can be fulfilled by utilizing several efficient protocols. Typically, B5G-IIoT applications require a throughput in Gbps. For efficient transmission, long-range IIoT devices with long-life batteries must be used.
- **Network topology:** The number of devices connected to other devices, as well as to cloud and fog architectures, must be carefully determined for improved QoS. Furthermore, network topology has a significant impact on QoS performance.
- **Security & safety:** Nowadays, security and privacy are key concerns. To avoid the leakage of confidential information or data loss, data collection and processing between inter- and intra-B5G-IIoT applications require strong privacy and security.
- **Reliability:** Many B5G-IIoT applications require ultra-high reliability. For example, in a smart healthcare system, it is critical to deliver data reliably. Furthermore, the QoS requirements of URLLC in B5G-IIoT demand high reliability in a variety of real-time applications.
- **Cost effective:** Low-cost B5G-IIoT smart devices and applications must be installed and used to reduce capital expenditure (CAPEX)/operating expenditures (OPEX). Deployment of applications such as smart factories and industries must satisfy market demands.
- **Standardization:** In IIoT, the many B5G standards are employed. These standards' implementation must be flexible and compatible with IIoT devices.
- **Device maintenance:** In an industry 4.0 environment, heterogeneous devices require constant management due to their connectivity to one another and to the Internet. SDN is used to describe such failures and the evolution of IIoT device maintenance issues.
- **Monitoring network:** Mobility, congestion, and overload of IIoT devices within a wireless network can all affect the network topology. As a result, continual and frequent

monitoring of the system is unavoidable. The number of smart devices in a IIoT application will grow over time. Thus, to avoid traffic congestion and overload, the system's settings should be capable of being adjusted in response to the traffic load and data requirements.

- **Configuration & system management:** It is critical to empower devices with self-configuration and control capabilities, as well as to reconfigure the network to accommodate new devices.
- **Scalability and integration:** Scalability poses various problems, such as how many smart devices are required to support an industrial application environment, how many devices can be comfortably handled by a server, and how to design a system efficiently within energy/spectrum constraints. Additionally, the integration of hybrid SDN and IIoT devices must be seamless and optimal.
- **Heterogeneity & interoperability:** Heterogeneous intelligent IIoT devices must communicate and exchange data with other devices, as well as share it through the Internet. This integration raises some serious challenges that must be addressed. Furthermore, standardization is desired for IIoT device interoperability.

### III. ENABLING AND EXISTING TECHNIQUES FOR 5G-IIoT

IIoT networks have a wide range of applications in electronic devices and industrial equipment, and are connected to billions of people. These numerous devices support a variety of network protocols and communication technologies; additionally, they are equipped with a variety of data processing and storage units and operate at a range of power levels. Each device is constrained by QoSs, energy efficiencies (EEs), and spectral efficiencies (SEs)/throughput, cost, power, security, reliability, and latency/delay. Industrial communication technologies are classified into wired and wireless categories according to their transmission procedures, and account for 85% of the global market [54]–[57]. IIoT communication technologies are illustrated in Fig. 5.

#### A. Wired Communication Technologies

Wired technologies include industrial field-bus and industrial Ethernet. These technologies provide reliable communication between the floor's control system and the higher-level control system. For cross-region communication between



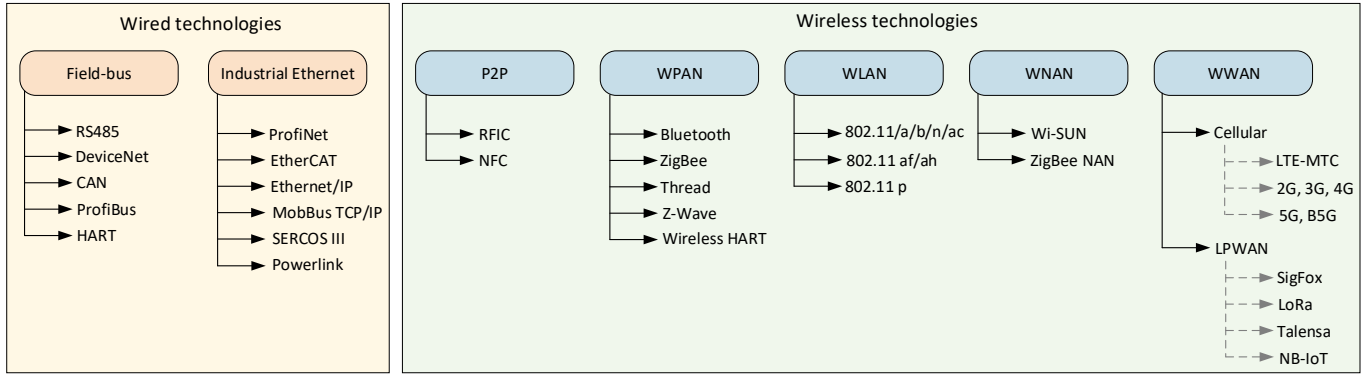


Fig. 5: IIoT communication technologies.

floors, control systems, and data centers, traditional Ethernet with its complex line topology is used, resulting in significant delays and inadequate security.

1) *Field-bus*: Field-bus communication has been used to connect actuators, sensors, and instruments. Moreover, data was transferred through field bus between these devices and higher control systems. It is the most extensively used wired industrial communication technology, consisting of RS485, DeviceNet, controller area network (CAN), process field bus (ProfiBus), and highway addressable remote transducer (HART).

2) *Industrial Ethernet*: For IIoT, industrial Ethernet is a suitable wired technology. When a huge amount of data needs to be conveyed, it gives a greater bandwidth than field bus technology. Ethernet protocols are becoming more prevalent as control system technologies evolve. The manufacturing execution system (MES) and supervisory control and data acquisition (SCADA) are examples of these control systems. Industrial Ethernet protocols include process field net (ProfiNet), ethernet for control automation technology (EtherCAT), Ethernet/internet protocol (IP), modbus transmission control protocol (TCP/IP), serial real-time communication system (SERCOS)-III, and Power-Links.

### B. Wireless Communication Technologies

There are numerous wireless technologies that can be used in IIoT. These technologies are classified into long and short range technologies based on their coverage range. Fig. 5 illustrates the major wireless technologies. The technologies are further categorised according to the services they offer, the requirements they fulfill, and the characteristics they possess [41], [58]–[63]. Table II summarizes the major parameters of the existing wireless technologies employed in IIoT. These technologies are described briefly in the following section.

1) *Peer to peer (P2P)*: P2P technology promotes the implementation of a cost-effective real-time infrastructure for IIoT that is both secure and low-latency [64], [65].

- **Radio-frequency identification (RFID)** is a wireless technology that is well suited for real-time, short-range, low-power, and low-cost communication in IIoT. It is used to transport small volumes of data between servers and devices in a short period of time.

- **Near field communication (NFC)** is a contactless data exchange high frequency RFID technology with low power consumption that is used to accomplish the shortest range possible. NFC is used to authenticate cards and facilitates server-to-server communication via P2P.

2) *Wireless personal area network (WPAN)*: WPAN has a range of up to 10 to 100 meters and is used to connect low-power devices that are centrally located [66], [67]. WPAN is a collection of low-cost devices such as bluetooth, zonal intercommunication global-standard (ZigBEE), Thread, Z-wave, and Wireless HART.

- **Bluetooth** is classified as WPAN and is widely regarded as the most appropriate technology for IIoT applications requiring low power. Initially, it supported only P2P communication, but was later expanded to support multi-point communication between up to 7 nodes for consumers.
- **ZigBEE** is also referred to as IEEE 802.15.4, is a low-power, short-range WPAN technology. It is optimal for medium-range IIoT applications that require a higher data rate. Home automation and sensor networks are two examples of IIoT applications.
- **Thread** is WPAN technology which is based on internet protocol version 6 (IPv6) low power mesh networking protocol. It is well suited for IIoT applications that emphasize functional efficiency, automation, and security. It is designed with the challenging features of low power consumption, a simple architecture, and enhanced security with no single point of failure. It is used for device-to-device and device-to-cloud communication and supports up to 250 nodes.
- **Z-Wave** is categorized as WPAN and is primarily used for IIoT home control and automation applications. Z-wave is a low-cost short-range network with a 200ms latency. It is capable of supporting up to 232 devices via a mesh network and operates flawlessly without any coordination. However, it performs optimally with 30-50 nodes, according to the manufacturer.
- **Wireless HART** communication is a two-way WPAN technology. It is designed to meet self-organizing and self-healing requirements using existing HART technologies. These are the requirements for flexibility, reliability, and security. It is optimal for IIoT applications that

TABLE II: Wireless technologies used in IIoT.

| Technology |               |         | Range (m)                                    | Frequency                                           | Max. Bandwidth      | Data Rate          | Power            | Cost |
|------------|---------------|---------|----------------------------------------------|-----------------------------------------------------|---------------------|--------------------|------------------|------|
| P2P        | RFID          |         | 10 m                                         | 13.56 MHz                                           | 1 MHz               | 640 Kbps           | Low              | Low  |
|            | NFC           |         | 20 m                                         | 13.56 MHz                                           | 1 MHz               | 848 Kbps           |                  |      |
| WPAN       | Bluetooth     |         | 60 m                                         | 2.402-2.480 GHz                                     | 2 MHz               | 1 Mbps             | Medium           | Low  |
|            | ZigBee        |         | 140 m                                        | ISM <2.4 GHz                                        | 5 MHz               | 40 Kbps, 250 Kbps  | Low              |      |
|            | Thread        |         | 140 m                                        | ISM <2.4 GHz                                        | 5 MHz               | 40 Kbps, 250 Kbps  |                  |      |
|            | Z-Wave        |         | 100 m                                        | 868.42 MHz - 908.42 MHz                             | 200 KHz             | 9.8-100 Kbps       |                  |      |
|            | Wireless-Hart |         | 225 m                                        | 2405-2480 GHz                                       | 5 MHz               | 250 Kbps           |                  |      |
| WLAN       | 802.11        | a       | Indoor 35 m,<br>Outdoor 120 m                | 2.4 GHz                                             | 20 MHz              | 1.5-54 Mbps        | Low              | Low  |
|            |               | b       | Indoor 35 m,<br>Outdoor 140 m                | 2.4 GHz                                             | 20 MHz              | 1-11 Mbps          |                  |      |
|            |               | n       | Indoor 70 m,<br>Outdoor 250 m                | 2.4-5 GHz                                           | 20-40 MHz           | 72-600 Mbps        |                  |      |
|            |               | ac      | 100-1000 m                                   | 5 GHz                                               | 20,40,60,80 MHz     | 433-6933 Mbps      |                  |      |
|            |               | af      | 1000 m                                       | 0.054-0.799 GHz                                     | 6-8 MHz             | < 568.9 Mbps       |                  |      |
|            |               | ah      |                                              | Sub GHz                                             | 1-16 MHz            | 150 kbps - 78 Mbps |                  |      |
|            |               | p       |                                              | Sub GHz                                             | 5/10/15 MHz         | 1.5-54 Mbps        |                  |      |
| WNAN       | Wi-SUN        |         | 1000 m                                       | ISM < 2.4 GHz                                       | 200 KHz - 1.2 MHz   | 50 Kbps - 1 Mbps   | Low              | Low  |
|            | ZigBee-NaN    |         | 5-10 km                                      | 868 MHz Europe, 915 MHz USA, 2.4 GHZ ISM World Wide | -                   | 200 Kbps           |                  |      |
| WWAN       | LPWAN         | SigFox  | 30 miles rural area,<br>2-6 miles urban area | 868 MHz Europe, 915 MHz North America, 433 MHz Asia | 100 Hz              | 100 bps            | Low              | Low  |
|            |               | LoRa    | 15-20 km                                     | 868 MHz Europe, 915 MHz North America, 433 MHz Asia | 250 KHz and 125 KHz | 50 Kbps            | Device dependent |      |
|            |               | Telensa | 2 km urban area, 4 km rural area             | Sub GHz                                             | -                   | 62.5-500 bps       | Low              |      |
|            |               | NB-IoT  | 1 km urban area, 10 km rural area            | Licensed LTE                                        | 200 KHz             | 200 Kbps           |                  |      |
|            | Cellular      | LTE-MTC | < 100 km                                     | 4G-LTE Band                                         | 20 MHz              | 10 Mbps            | PSM              | High |
|            |               | 4G      | < 100 km                                     | LTE Band                                            | 5-20 MHz            | 100 Mbps           | High             |      |
|            |               | 5G      | < 100 km                                     | Sub 6 GHz, mmWave                                   | 100-200 MHz         | 10 Gbps            | High             |      |

require factory automation and monitoring of mission-critical equipment.

3) *Wireless local area network (WLAN)*: WLAN has a coverage area of between 100 and 1000 meters, which typically extends from a floor to a building. WLAN transmits data via radio waves; however, cables are now used to connect the network's backbone. IEEE 802.11 a/b/n/ac, 802.11 af, 802.11 ah, and 802.11p are among the technologies supported by WLAN.

- **IEEE 802.11 a/b/n/ac** standards supported by the routers are ideal for connecting multiple devices on-site while maintaining network security in an IIoT network. These technologies enable both indoor and outdoor dual band radio access points (APs) that are capable of receiving and transmitting in both bands. Moreover, because these technologies operate at low frequencies, they can easily obstruct concrete structures, such as walls and buildings.
- **IEEE 802.11 af** commonly known as White-Fi, is classified as WLANs. It is applicable throughout a broad spectrum, depending on the application. White-Fi consumes less power and is used to connect devices across a large geographical region where low-speed connections are tolerable.
- **IEEE 802.11ah** is another WLANs technology, and it is referred to as WiFis Halow or a low-power WiFis. It is employed in a wide range of IIoT applications, including

gadgets, appliances, smart sensors, and actuators, both indoors and outdoors. Additionally, it is used in both urban and rural settings. Its popularity is increased by the backhaul aggregation and hotspot. IEEE 802.11p is a member of the WLANs family of standards, which enables increased data rates/bandwidth with reduced latency/delay. It is optimal for IIoT applications such as smart cities and networking.

4) *Wireless neighbourhood area network (WNAN)*: WNAN is a self-organizing and self-healing mobile ad-hoc network. The low cost, low-power RF circuits with multi-channel transmitter and receiver are the key attributes of WLAN, making it ideal for long-range IIoT applications.

- **Wireless smart ubiquitous network (Wi-SUN)** is an exciting technology for interconnecting sensors, data centers, and bidirectional control systems in a IIoT network. Additionally, it is a self-organizing, self-healing, and flexible technology that utilizes mesh networks to construct dense networks capable of supporting many connections. Additionally, it consumes less energy and offers enhanced security.
- **ZigBEE neighbourhood area network (NAN)** enables low-power wireless networks to achieve a high level of network functionality and self-healing capability. Additionally, it is extremely reliable, robust, and scalable. The technology is ideal for smart energy, health care, and

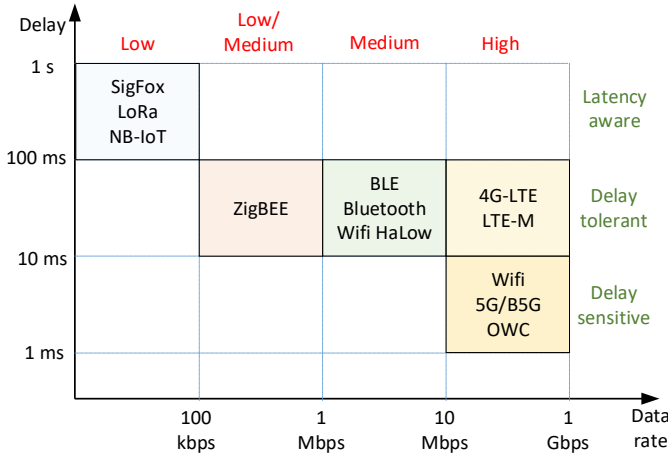


Fig. 6: Data rate vs. latency for various 5G-IIoT communication technologies.

factory/home automation applications.

5) *Wireless wide area network (WWAN)*: WWAN covers an area of up to 100 kilometers. In a WWAN, short packets are transmitted between sensors. It encompasses cellular network technologies such as LTE-MTC, second generation (2G), third generation (3G), 4G, 5G, and B5G, as well as LPWAN technologies such as SigFox, LoRa, Telensa, and NB-IoT.

a) *Cellular Networks*: The IIoT can communicate over long distances by utilizing an already existing cellular network. Among these cellular technologies are global system for mobile communications (GSM), 3G, and 4G. However, because 4G is capable of transmitting data over a long distance at a high data rate, it consumes more power. It can be implemented using sensor-based low-bandwidth devices, such as a Raspberry-pi or an Arduino, that are capable of transmitting small amounts of data over the Internet [21], [42]. Recent cellular technologies enable the provision of more sophisticated services for achieving IIoT targets. Several critical cellular technologies for IIoT are discussed in this section.

- **4G** standard offers support for long range and high data rate communication as compared to WiFi, Bluetooth, White-fi, and HaLow for IIoT applications. 4G provides better coverage than 3G, while 3G provides better as compared to 2G. On the other hand, 2G consumes approximately 50% less power than 3G, while 3G consumes approximately 50% less power than 4G. These cellular standards enable affordable coverage and data rates in a limited number of IIoT applications, such as mines, warehouses, and manufacturing industries.
- **LTE-MTC** is a low-power wide area network technology developed by 3GPP and classified as WWAN. LTE-MTC provides a higher data rate with increased mobility and battery life at the expense of increased bandwidth. LTE-MTC is typically used in IIoT applications that require low power, increased coverage, and minimal complexity. Among these applications are smart grid and future machine to machine (M2M) networks. The technology operates under power saving mode (PSM).

- **5G/6G** key objectives are largely consistent with the IIoT requirements and are as follows: 1) URLLC: delivering high reliability and low latency to time-critical, real-time mission-critical applications such as disaster management. 2) eMBB: significantly increase the bandwidth and speed of smart city applications. 3) glsmmtc: connects millions of devices over long distances [61], [68]–[71]. Thus, the key objectives of B5G-IIoT are summarized as follows:

- URLLC, for real-time communication.
- eMBB and mMTC are used to increase the bandwidth and connectivity of a large number of devices.
- Security through network slicing.
- Edge computing is used to ensure proper edge functionality and to share local computations with edge devices.

b) *LPWAN*: The prime objective of 5G/B5G-IIoT is to provide a long-range, cost-effective communication infrastructure with minimal power consumption. LPWAN is an intelligent network option for connecting multiple IIoT applications. The advantageous features of LPWAN, such as long range communication, increased coverage, low power consumption, and increased throughput/data rate, make it an acceptable solution for IIoT. It is a critical technology for the implementation of B5G-IIoT [26], [53], [72]–[75].

- **SigFox** widely considered to be the leading international IIoT technology because it enables massive connections to millions of devices and data transfer without the need for a predefined network [49], [76], [77]. It is a widely used wireless technology that bridges the gap between WiFi and cellular networks. Additionally, it significantly improves EE via software-assisted communication. As a ultra narrow band (UNB) technology, it supports low data rates. As a result, it provides a reliable, energy-efficient network solution for connecting millions of devices spread across a large geographical area. Typically, this area spans several kilometers, making SigFox ideal for IIoT applications such as smart meters, environmental sensors, and security devices.
- **LoRa** is another well-known B5G-IIoT technology that is listed under LPWAN. It is used to transmit secure data over long distances of up to several kilometers while using little power. Numerous IIoT networks utilize LoRa to provide services over a larger area than cellular networks. It operates at a wide range of frequencies between 868 MHz and 915 MHz. LoRa integrates easily with existing network infrastructure that requires low-power IIoT applications. It is well suited for IIoT applications such as smart cities that require broad coverage with low power consumption, low cost, and bidirectional communication [41], [58].
- **Telensa** is an LPWAN technology that enables the deployment of sensors and actuators as a hardware platform for software-defined services. It is suitable for IIoT applications such as smart cities, where smart street lights are regarded as low-cost and low-power sensors on a global scale.



- **NB-IoT** is an LPWAN subtype of WWAN that was introduced with the 3GPP release 13 standardization. NB-IoT develops NR technology for IIoT applications such as smart home, smart city, and security systems. Additionally, due to its low power consumption characteristics, it offers promising solutions for future generation wireless technologies. It is a complementary solution to the B5G, URLLC, and mMTC requirements, which are required for the IIoT attributes.

#### IV. STANDARDIZATION OF URLLC AND EMBB IN 5G-IIoT

IIoT requires stringent QoS requirements in both wired and wireless environments, including throughput, latency, reliability, coverage, power efficiency, and mobility [140], [141]. The enabling features and services provided by 5G/B5G, such as URLLC, eMBB, and mMTC, can assist IIoT in achieving the aforementioned QoS requirements. As a result, B5G-IIoT is regarded as an optimistic solution. As a result, we classify technologies according to the QoS requirements of IIoT in URLLC and eMBB. This section discusses technologies in terms of standardization.

Table III summarizes the key enabling communication technologies. The key enabling technologies in B5G-IIoT which have been aided by URLLC are multi-RAT, 4G, LTE, 5G/B5G, and V2X. Multi-RAT and CoMP, 4G, and LTE are all included in the B5G standard as latency reduction techniques for URLLC. These techniques are based primarily on mm-Wave communication, ultra dense small cell network (UDN), and 3GPP Release 16 in device to device (D2D). Similarly, V2X is an application-oriented communication technique that includes vehicle-to-vehicle (V2V), vehicle-to-network (V2N), and vehicle-to-infrastructure (V2I) communications. V2X adheres to multiple protocols to reduce latency and increase reliability, as well as to ensure compatibility with mission-critical IIoT applications. Similarly, 3GPP Release 15 and 16 employ a range of licensed and unlicensed spectrum techniques, such as LPWAN and WWAN, to enable the use of B5G-NR, UDN, 4G-LTE, and 5G new waveforms to deliver attractive multi-Gigabit data rates and multi-point connectivity to industry end consumers.

B5G-NR specifies three primary objectives, namely increased network density, increased SE, and access to high spectrum bands. Nonetheless, eMBB integrates multiple techniques, such as B5G-NR communication with UDN, mm-Wave, and FD communication. mm-Wave communication supports a wide variety of IIoT applications, including time-sensitive ones, with a data rate of up to Gbps. In Fig. 6, we compare various B5G-IIoT communication technologies in terms of their data rate and latency requirements. 5G/B5G-IIoT aims to provide industrial infrastructure with flexibility and scalability. As a result of the extremely low latency, high reliability, and high throughput of IIoT, it is more effective in automation control and monitoring systems. Following that, this section discusses technologies and their standardization. The key technologies in 5G standardization that can assist in 5G-IIoT compatibility are illustrated in Fig. 7.

##### A. 3GPP Release 15

In Release 15, the first phase of the 5G system is referred to as New Radio, or NR. With this 5G-NR standard, eMBB and eMBB can operate at frequencies up to 52.6 GHz. Release 15 of RAT release will include support for both non-standalone (NSA) and standalone (SA) connectivity. The Release 15 system will include critical features such as network slicing, access and mobility management, a QoS framework, a policy framework, and network sharing, as well as the interoperability of unauthorized 3GPP networks, i.e. WiMAX, WiFi, and integration with evolved packet system (EPS). The 5G system architecture and procedures will be described in detail in Release 15 specifications.

There is discussion of 3GPP Release 15, which established the basic framework for URLLC in NR, primarily in support of IIoT. Powered by B5G networks, the IIoT enables new value creation and significantly increases revenue opportunities for original equipment manufacturers (OEMs). IIoT will improve efficiency through industrial automation, while also utilizing a highly flexible, scalable, and virtualized infrastructure. URLLC is based on Release 15 of the 3GPP specification. Two critical requirements for URLLC have been specified in 3GPP:

- Target user plane latency of 0.5 ms for the downlink (DL) and uplink (UL).
- Packet accuracy of 99.999% for a 32-byte packet with a 1 ms user plane latency.

Satisfying URLLC specifications is extremely difficult due to the coupling of high accuracy and low latency. To accomplish this, 3GPP Release 15 defines two distinct physical layer (PHY) feature groups for URLLC support.

The first group contains PHY features that are used to reduce latency. Additionally, features such as flexible slot structure, flexible scheduling, and fast are included. In the second group, PHY features are used to increase reliability. The following features pertain to data enhancements:

- To improve reliability, identical data packets are transmitted from multiple transmission points. This is accomplished through the use of the packet data convergence protocol layer, which duplicates the data packet. It can be used to send the same packets to the user equipment (UE) from two different next generation nodeBs (gNBs) to improve reliability. Repetition with multiple data channel slots – Two, four, or eight repetitions can be used to increase reliability.
- The URLLC enables extremely high reliability through the use of special modulation and coding values.

The major characteristics of NR Release 15, which enables eMBB (10 Gbps - 20 Gbps), are as follows: Ultra-wide bandwidth (up to 100 MHz in  $< 6$  GHz, up to 400 MHz in  $> 6$  GHz), multiple numerologies for optimal operation at various frequencies, native forward compatibility mechanisms, new channel coding, native support for low latency and ultra reliability, flexible and modular RAN architecture that supports split fronthaul, split control- and user-plane, and native end-to-end network slicing support.

TABLE III: Enabling technologies for URLLC and eMBB in 5G-IIoT.

| Services for 5G/B5G-IIoT | Enabling Technologies                                                                                               | Communication Standards                   | References                     |
|--------------------------|---------------------------------------------------------------------------------------------------------------------|-------------------------------------------|--------------------------------|
| URLLC                    | vehicle-to-everything (V2X) communication                                                                           | 5G/B5G, NB-IoT                            | [78]–[90]                      |
|                          | Multi-radio access technology (RAT): coordinated multi-point transmission and reception (CoMP), 4G-LTE, WiFi, 5G NR | LTE-M, WiFi/WiFi Halow, NB-IoT            | [37], [51], [91]–[97]          |
|                          | Instant and reserved scheduling                                                                                     | LoRa, NB-IoT, cellular                    | [34], [98]                     |
|                          | 5G/B5G, 5G/B5G-NR                                                                                                   | Cellular, LPWAN                           | [99]–[103]                     |
|                          | Packet and frame structure                                                                                          | LoRa, B5G                                 | [51], [51], [104]–[110]        |
| eMBB                     | B5G, B5G-NR                                                                                                         | NB-IoT, LoRa                              | [111]–[117]                    |
|                          | Flexible data-rate (FD) communications                                                                              | WWAN, WiFi/WiFi Halow, Bluetooth, ZigBee  | [112], [118]–[122]             |
|                          | NW techniques                                                                                                       | B5G                                       | [25], [51], [106], [123]–[129] |
|                          | Unlicensed spectrum based techniques                                                                                | LTE-M, NB-IoT, LoRa, Talensa, Sigfox      | [28], [29]                     |
|                          | Dense small cell network                                                                                            | WWAN, WiFi/ WiFi Halow, Bluetooth, ZigBee | [130]–[134]                    |
|                          | millimeter wave (mm-Wave)                                                                                           | B5G                                       | [51], [111], [135]–[139]       |

### B. 5G Evolution: 3GPP Release 16

The enhancements to 5G-NR in 3GPP release 16 include both extensions to existing features and new verticals and deployment scenarios. Unlicensed spectrum operation, intelligent transportation systems, IIoT, and extraterrestrial networks are a few of the highlights [142], [143].

Among them is the capabilities of 5G to support a large number of mobile and fixed IIoT devices with varying speeds, bandwidths, and QoS requirements. With the expansion of IIoT, the flexibility of 5G will become even more critical for enterprises that require support for the stringent requirements of critical communications. Due to the ultra-reliability and low latency of B5G, self-driving cars, smart energy grids, enhanced factory automation, and other demanding applications will be possible. As 5G improves network performance, cloud computing, artificial intelligence (AI), and edge computing all contribute to the handling of the data volumes generated by the IIoT. Additional 5G enhancements, such as network slicing, private networks, and 5G core, will ultimately contribute to the realization of the vision of a IIoT network capable of supporting a massive number of connected devices. The most significant enhancements in release 16 are in the areas of multiple-input multiple-output (MIMO) [144], [145] and beamforming enhancements, dynamic spectrum sharing (DSS) [146], dual connectivity (DC) and carrier aggregation (CA) [147], and power savings in UE [148], [149].

This release is targeted at industrial automation use cases. The 5G-URLLC foundation has been enhanced in Release 16 to ensure more reliable communication of up to 99.99%. Increased retransmissions are insufficient for these use cases, as there is a strict latency constraint as well. CoMP can be used in conjunction with other technologies to overcome this

constraint. Multiple transmission and reception architecture (Multi-TRP) establishes redundant communication paths with spatial diversity, ensuring that communication is maintained even if one of the paths is temporarily blocked by using the remaining paths [32], [33]. Moreover, this release addresses new vertical and deployment scenarios, such as integrated access and backhaul (IAB) NR in unlicensed spectrum features related to IIoT and URLLC.

The IIoT is a significant vertical focus of NR Release 16. It includes latency and reliability enhancements on the already existent Release 15's work on high reliability and low air-interface latency. This release expanded IIoT use cases such as factory automation and electrical power distribution. Additionally, there is support for time-sensitive networking (TSN), which is critical for time synchronization. When the URLLC-related improvements are added together, they have a significant effect on NR. Release 16 extends the inter-UE DL preemption support introduced in Release 15 to the UL, allowing a UE to cancel a lower-priority UL transmission in favor of a higher-priority UL transmission from another UE. Additionally, Release 16 supports standardizing the resolution of intra-UE UL resource conflicts. Control-channel monitoring is more frequent in this release to reduce latency. Moreover, multiple configurations for both UL configured grants and DL semi-persistent scheduling can be active concurrently. The enhancements are especially beneficial for TSN traffic, where the base station (BS) is aware of the traffic pattern [150].

### C. 5G Evolution: 3GPP Release 17

The 3GPP release 17 includes support for new use cases, such as public safety, non-terrestrial networks, and private networks, as well as enhancements to existing ones, such

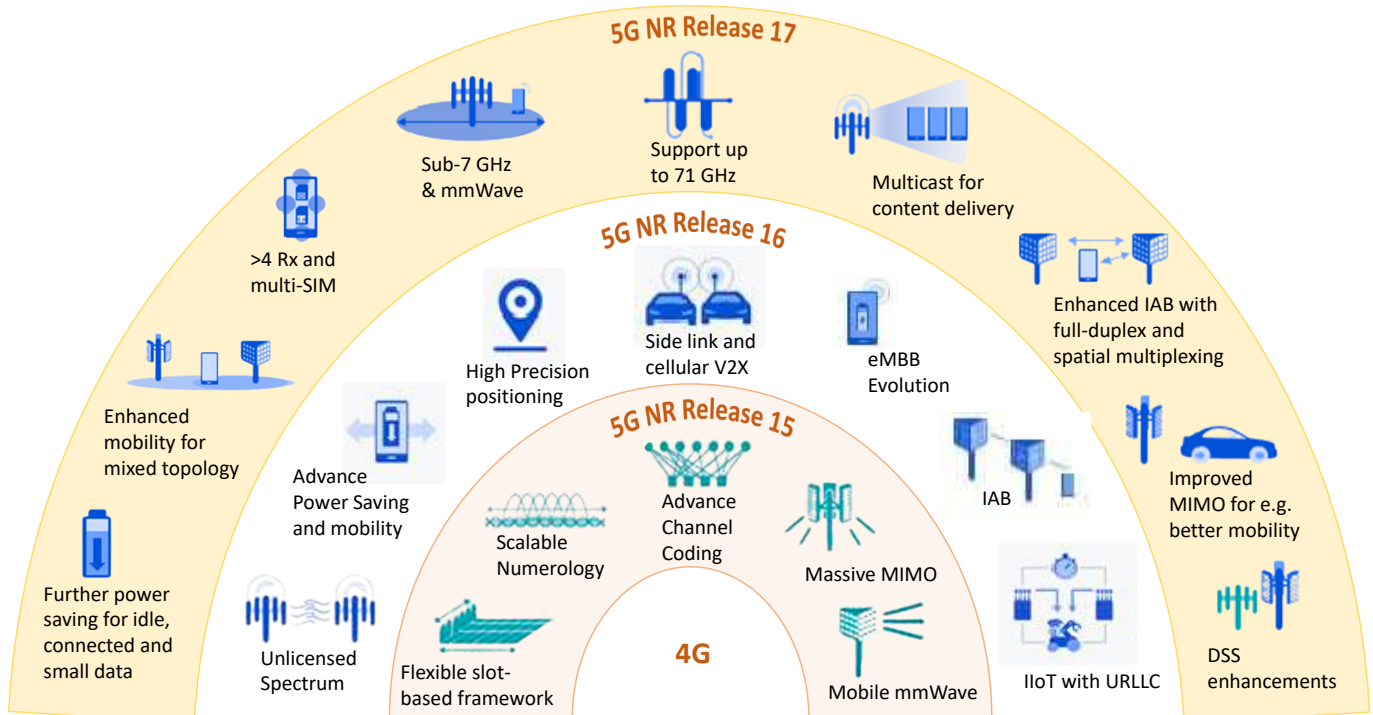


Fig. 7: Techniques in 5G standardization that can assist 5G-IIoT.

as mobile broadband, industrial automation, and vehicle-to-everything. Moreover, this release includes improvements to existing services and use cases, as well as new deployment alternatives and use cases.

Together with previously established use cases and deployment options, 5G will incorporate intelligent network technologies and support a large number of new use cases. One of the important components of 5G is the application of AI based on machine learning (ML) techniques. The implementation of AI/ML in future wireless networks will trigger a paradigm shift. AI and ML will be used to solve multi-dimensional network optimization problems in an online or offline manner, enabling the introduction of intelligent network management. By optimizing the performance of multi-antenna systems with AI/ML, for example, the radio interface can be enhanced as well. Wireless networks will enable immersive experiences in virtual and physical environments, as well as human-machine interaction with devices and wearables in IIoT, through extended reality communication.

A significant portion of the functionality already implemented in live NR networks will be enhanced in Release 17, which mark the beginning of the transition to 5G Advanced or B5G.

1) *Beamforming and MIMO*: Enhancements in Release 17 MIMO focus on four areas: beam management; Multi-TRPs to enable URLLC; Multi-TRPs to enable eMBB; and reciprocity in time division duplex (TDD) and frequency division duplex (FDD).

The goal of streamlining signaling and optimizing operator equipment (UE) with multiple antenna panels is to increase performance while maintaining high mobility. Multi-TRP has been added to the physical downlink control channel (PD-

CCH), physical uplink shared channel (PUSCH), and physical uplink control channel (PUCCH). Along with providing more detailed channel state information (CSI) feedback for non-coherent joint transmission, the new algorithms improve performance for high-speed train communications. Enhancements to reciprocity-based operation include the creation of new codebooks with reduced feedback overhead, the availability of channel information at gNB, and enhancements to the sounding reference signals.

2) *Dynamic Spectrum Sharing*: Using the DSS included in Release 15, it is now possible to deploy both a LTE and a NR cell from the same BS. Release 16 primarily increased the capacity of the physical downlink shared channel (PDSCH). Moreover, the release enables operators to better manage PDCCH resource shortages in the NR cell that occur as the number of NR UEs increases. Cross-carrier scheduling, introduced in Release 17, schedules data channels on the shared primary cell via the PDCCH of a DL secondary cell.

3) *User Equipment Power Savings*: Release 17 includes power-saving enhancements for UEs operating in connected, idle, or inactive radio resource control (RRC) modes. Both eMBB UEs and RedCap devices are specified with power-efficiency enhancements. Several power-saving enhancements are included in this release's features, including relaxed radio resource monitoring for devices operating at low mobility or under ideal radio conditions, extended discontinuous reception for devices with a low latency tolerance, reduced PDCCH monitoring during active time, and power-efficient paging.

4) *Positioning*: NR added support for positioning via LTE (for non-standalone deployments) and RAT-independent positioning (Bluetooth, wireless LAN, pressure sensors, etc) in Release 15. A new version of Release 16 introduced time-

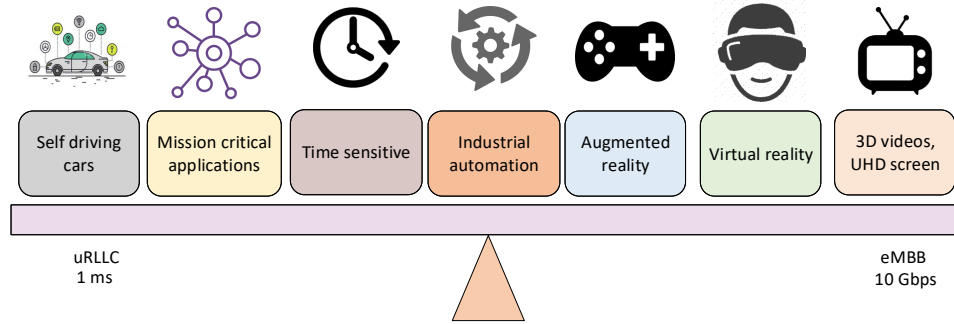


Fig. 8: Trade-off between URLLC and eMBB.

based positioning methods for NR standalone deployments (multi round trip time (RTT), the DL time difference of arrival, and the UL angle of arrival), which can be used in conjunction with timing-based solutions to improve accuracy.

Release 17 further improves NR positioning for specific use cases such as factory automation by aiming for 20-30 cm accuracy in certain deployments. Additionally, Release 17 improves latency reduction, enabling positioning to be accomplished in time-critical applications such as remote control. Along with high-accuracy positioning, the integrity of location information is critical in IIoT and automotive use cases. The purpose of this release is to introduce key performance indicators to demonstrate the reliability/integrity of a measurement report that addresses the global navigation satellite system (GNSS) positioning procedure.

5) *Ultra Reliable Low-Latency Communication*: Using URLLC, 5G has penetrated into multiple vertical markets. Release 15 established a strong foundation, and Release 16 added enhancements by the 3GPP's system architecture and RAN groups to improve support for vertical industries such as factory automation, shipping and transportation, and electrical power distribution. Several redundancy schemes were added to the user plane as part of these enhancements, as well as improvements to reliability, latency reduction, and support for time-sensitive communications (TSC). Enhancements in Release 17 are intended to increase SE and system capacity, while also supporting URLLC in unlicensed spectrum environments and strengthening the framework for supporting TSC. Additionally, enhancements to hybrid automatic repeat request (HARQ), CSI, intra-UE multiplexing, and service survival time will be included in the assistance information provided by the TSC.

6) *NR coverage*: The coverage of a company or organization has a direct impact on service quality, operating expenses, and capital expenditures, making it a critical factor in commercialization and competition. The PUSCH is identified as a potential coverage bottleneck in Release 17. As part of the 3GPP initiative to expand PUSCH coverage, mechanisms for transport block processing repetition and support for multiple slots are being investigated. Additionally, Release 17 specifies mechanisms for grouping demodulation reference signals (DMRS) and indicating dynamic PUCCH repetition factors.

7) *Small Data Transmission*: The existing NR RRC inactive mode promotes power-efficient connection establishment

by allowing UEs to resume a previously established RRC connection. According to Release 17, data transmission in RRC inactive mode further reduces the power consumption of UEs during system access. This is especially important for low-power devices that transmit data infrequently and in small amounts in order to minimize control plane signaling overhead by avoiding the need to resume an RRC connection.

8) *Non-Public Networks*: 3GPP Release 16 defined non-public networks (NPN) as those that provide access to a subset of users, such as devices belonging to a particular factory. The 3GPP specified two options for NPN deployment as part of its support for industrial verticals. An NPN that is integrated into the public network enables operators to support NPNs by directly integrating them into their networks. A standalone NPN is another option for deployment. As a general rule, a standalone NPN is functionally and characteristically comparable to a standard public network. Standalone NPN receive additional enhancements in 3GPP Release 17. The enhancements include the ability for a UE to attempt access to a standalone NPN using external credentials (such as those from a public network or another standalone NPN), UE onboarding to the standalone NPN (for example, by providing it with NPN credentials and/or subscription parameters), and support for emergency services.

9) *Edge Computing*: Release 15 of 3GPP enables operators and third parties to host services close to the UE's access point of attachment. The baseline architecture reduces end-to-end latency and the load on the transport network. In Release 17, mechanisms for discovering edge application servers are introduced. For instance, it defines an edge application server discovery function (EASDF) whose primary purpose is to facilitate the session breakout connectivity method. In addition to acting as a domain name system (DNS) resolver, the EASDF can respond to DNS queries with UE location information. The UE locations can then be resolved to nearby application servers via the DNS.

Release 17 clarifies and enhances the use of user-equipment route-selection policy (URSP) rules in the distributed anchors and multiple protocol data unit (PDU) session connectivity models. When configuring URSP rules in the UE, application server information can be taken into account. As a result, the UE can establish PDU sessions dynamically for specific application servers, obviating the need for complex session breakout solutions. Additionally, Release 17 introduces new



mechanisms for exposing QoS monitoring results and enhances support for application server relocation in the event of a UE's mobility.

## V. TRADE-OFF BETWEEN URLLC AND EMBB FOR 5Gs/B5G-IIoT

B5G-IIoT features various use cases related to high definition (HD)-video, virtual reality (VR), augmented reality (AR), smart cities, smart cars, smart factories, smart grid, remote control and autonomous robots and AI networks. These use cases possess sensors, actuators and processors along with massive IIoT applications. However, EE, SE, latency, reliability and throughput requirements vary for different IIoT applications. ITU has defined the QoS requirements for URLLC, mMTC, and eMBB in B5G standard. For URLLC, a reliability of 99.99% and less than 1 ms latency is required. EE and connection of massive IIoT devices is a parameter for mMTC, and throughput of up to Gbps is desired for applications categorized under eMBB. Therefore, B5G provides effective services to the IIoT applications and their corresponding use cases under varying QoS requirement. However, use cases which achieve URLLC suffers with a degradation of eMBB. On the contrary, if eMBB is achieved URLLC would be degraded. Therefore, a trade-off between the two is of particular emphasis, i.e., to achieve URLLC and eMBB simultaneously with same resource efficient method to satisfy the QoS requirement defined by ITU. Thereby, the coexistence of URLLC and eMBB open the doors to new technologies and research directions [31], [36], [174]–[178].

### A. Performance Trade-off between URLLC and eMBB

Design trade-off between URLLC and eMBB is widely investigated by the research community. Some key trade-offs determined by the researchers are discussed herein.

1) *Finite- vs. large-block length*: In the recent past, the study of the effective capacity of fading channels has caught the attention of many researchers in order to understand the performance of wireless systems under the influence of statistical queuing. Although channel coding is performed using a finite block of symbols, in almost all prior studies the service rates of the queuing model were assumed to be equal to the instantaneous capacity values. In addition, transmissions are assumed to be error-free with no decoding errors. It is important to note, however, that error-free communication at the rate of channel capacity can be achieved as code-word length increases without limit. In order to have high reliability or lower error probability, finite block length codes will necessarily use coding errors and will transmit at a lower rate than the channel capacity. However, due to noise, deep fading, collisions, and interference, low-latency applications with small block lengths are prone to failure. If  $k$  information bits are transmitted using coded packets over  $n$  channel uses, the maximum coding rate  $R = \frac{k}{n}$  is lower than the Shannon rate. Additionally, in order to ensure high reliability, encoded data must be transmitted at a lower rate than Shannon's capacity. With a large blocklength, no errors occur and the additive white gaussian noise (AWGN) channel's achievable

rate is equal to Shannon's known capacity  $R = \log_2(1 + \gamma)$ , where  $\gamma$  is the achieved signal to noise ratio (SNR). Shannon's capacity model has been shown to significantly overestimate delay performance for these applications, resulting in an insufficient allocation of resources. A theoretical framework is needed to model the performance of such systems due to short time spans, finite block lengths, and interference despite the huge interest in the field [179], [180].

2) *Spectral efficiency vs. latency*: Additionally, transmission schemes must be capable of supporting a significant number of devices within a given bandwidth while maximizing battery life. With this approach, effective bandwidth has the potential to provide a good balance between SE and transmission time for a given coverage. To avoid excessive transmission time, the allocated bandwidth for a device should not be much less than the effective bandwidth. To be able to accommodate more B5G-IIoT devices while utilizing the available bandwidth without significantly increasing transmission time, it is preferred to allocate bandwidth that is not greater than the device's effective bandwidth for a given sensitivity level. Due to the short transmission time interval (TTI), low latency incurs a SE penalty. SE and latency are not characterized in a multiple access and broadcast system that takes into account: a) bursty packet arrivals; b) traffic that is low latency and delay tolerant; and c) channel fading and multipath. To effectively support the huge number of B5G-IIoT devices in 5G and beyond cellular networks, advanced transmission scheduling techniques and low signaling overhead medium access control (MAC) protocols need to be investigated [181].

3) *Energy efficiency vs. latency*: An important trade-off that needs to be identified is energy consumption versus latency. The battery life of B5G-IIoT devices is extended by placing them in sleep mode or deep sleep mode while they are not in use. The device needs to check periodically for packets sent by applications on the network, since applications sending packets to the device may wake it up. In addition to the latency of the packets, device check frequency, determining how much energy is consumed. As the frequency increases, the latency will decrease, but energy consumption will rise. When reliability deteriorates, re-transmissions are necessary, which means more energy is consumed to provide a target level of reliability. Energy is measured generally using  $\sqrt{d^{-1}}$  or  $\sqrt{d^{-1}} \log d$ , where  $d$  is the delay. This means that the energy decreases quite slowly with increasing delay [182].

4) *Reliability vs. latency vs. data rate*: IIoT and next-generation wireless communication standards, including 5G and B5G, provide applications which requires rethinking the communication stack, practical codes, networking architecture, and control design to provide ultra-low latency of 1 ms, extremely high reliability of more than 99.99%, and 20 Gbps peak data rates. However, higher reliability comes at a cost of higher latency. Typically, the reliability is ensured with re-transmissions which induce additional delays. However, higher data rates is achieved at the expense of lower reliability, and vice versa.

5) *SNR vs. diversity*: Diversity is typically defined when plotted against the SNR on a logarithmic scale as the slope of probability of error. The key question is: how does the SNR

TABLE IV: URLLC and eMBB based B5G-IIoT applications; Lat.: Latency, Rel.: Reliability, Tp.: Throughput, Cell dens.: Cell density.

| Smart applications  | Use cases                                | Lat. (ms)  | Rel.                  | Tp. (Mbps) | Cell dens. (Gbps/km <sup>2</sup> ) | Ref.                            |
|---------------------|------------------------------------------|------------|-----------------------|------------|------------------------------------|---------------------------------|
| Factories           | Printing & packaging machines            | 2 & 1      | $10^{-9}$             | 1, < 100   | 10                                 | [19], [151]–[163]               |
|                     | Machine tools & manufacturing            | 0.5 & 5    | $10^{-9}$             | < 100      | < 10                               | [151], [156], [163]             |
|                     | Monitoring automation & remote control   | 50         | $10^{-3}$ – $10^{-5}$ | 1, < 100   | < 10                               | [19], [156]–[162], [162], [163] |
|                     | Remotely operated video & motion control | 10–100 & 1 | $10^{-9}$             | < 100      | 14                                 | [152]                           |
|                     | Industrial robots                        | 4–8        | $10^{-9}$             | < 100      | 10                                 | [164]–[167]                     |
|                     | Milling machines & cranes                | 4–8 & 12   | $10^{-9}$             | 1, < 100   | 10                                 | [164]–[167]                     |
| Autonomous vehicles | Accidental warnings                      | 10         | $10^{-3}$ – $10^{-5}$ | 0.1–29     | N/S                                | [79], [162], [168], [169]       |
|                     | High speed train                         | 10         | N/S                   | 25–50      | 12.5–25                            | [19], [154], [155]              |
|                     | Automated driving                        | 5          | $10^{-5}$             | 0.1–29     | N/S                                | [19], [154], [155]              |
| Grid Stations       | Medium electricity distribution          | 25         | $10^{-3}$             | 10         | 10                                 | [19], [154], [155]              |
|                     | High electricity distribution            | 5          | $10^{-6}$             | 10         | 100                                | [19], [154], [155]              |
| Cities              | Outdoor industrial events                | 1–2        | $10^{-2}$             | 30         | 900                                | [155], [162], [168]–[173]       |
|                     | Public gatherings                        | 10–100     | $10^{-2}$             | 0.3–20     | 100–10000                          | [155], [162], [168]–[173]       |
|                     | Shopping centers                         | 10–100     | $10^{-2}$             | 60–300     | 60                                 | [155], [162], [168]–[173]       |
|                     | Mass communication                       | 200–5000   | $10^{-2}$             | 15         | 60                                 | [155], [162], [168]–[173]       |
|                     | Urban area                               | 1–2        | $10^{-2}$             | 60–300     | 700                                | [155], [162], [168]–[173]       |
| Traffic management  | Traffic congestion                       | 8          | $10^{-3}$             | 20–100     | 480                                | [154], [155]                    |
|                     | Traffic management efficiency            | <100       | $10^{-3}$             | 10         | 10                                 | [19], [153]–[155]               |
|                     | Urban crossings                          | < 100      | $10^{-5}$             | 10         | 10                                 | [19], [153]–[155], [168]–[173]  |
|                     | Safety measures for highway              | 10–100     | $10^{-3}$ – $10^{-5}$ | 10         | 10                                 | [19], [153]–[155]               |
|                     | safety measures for urban area           | 10–100     | $10^{-3}$ – $10^{-5}$ | 10         | 10                                 | [19], [153]–[155]               |

requirement decrease with increased diversity and nodes in the network? How does the use of higher frequency bands affect diversity? It's important to know how much SNR is needed to balance time-varying channels, and bad fading events. SNR significantly effects the diversity but compensates the time varying channels.

6) *Open vs. closed loop*: In closed loop system low data rates are obtained due to the utilization of limited resources. In an open loop, the cooperative devices, like the relay nodes, broadcast the packets for transmission which results in a reduced delay. Moreover, the diversity puts a positive impact on the reliability. Multiplexing techniques, such as open-loop MIMO, maximize multiplexing gains. Despite this method's ability to transmit multiple data streams using multiple transmit antennas, it requires a complicated detection algorithm in the receiver. Unlike open-loop MIMO, closed-loop MIMO techniques build capacity or improve SNR based on channel knowledge. Since closed-loop MIMO in highly mobile environments experiences a delay between transmitting and receiving channel information, designers should be careful when using it. In addition, MIMO with closed-loop feedback and limited uplink sounding performs poorly due to incomplete channel knowledge.

### B. Enabling Technologies and Standards for Trade-off between URLLC and eMBB

Many existing enabling technologies cope with the trade-off between URLLC and eMBB. However, this is still an open research question to obtain an optimal trade-off point.

1) *TTI*: The TTI in 5G NR and LTE is reduced to 1 ms and 0.125 ms, respectively, by introducing wider spacing between sub-carriers. Further improvement is realized by reducing the RTT during HARQ retransmissions, eventually leads to a significant reduction in latency. To satisfy the QoS constraint of reliability, minimum time should be consumed during HARQ retransmissions. Having said that, the reduction in symbol duration and growing spaces between the sub-carriers also lessen the achieved rate and available resources for the transmission. Therefore, a trade-off is imperative between the latency and sub-carrier resources.

2) *Multiplexed eMBB and URLLC*: For an efficient and optimized system, the resource partitioning is static or semi-static between eMBB and URLLC. Typically, high frequency is required to achieve high degree of reliability besides low latency. However, intelligent and optimized scheduling schemes are used to transmit low latency packet, while relaxing the eMBB.

3) *Network slicing and edge caching*: Edge caching and cloud computing have shifted and shared the network load to the edges, thereby, reduced the latency of URLLC. In parallel, additional bandwidth and caching resources to resource are gained through network slicing. These techniques have provided a boost to the sensitive applications like AR and VR.

4) *ML & AI*: ML has gained an overwhelming attention of the scientific community during the last decade. Typically, the ML executes the global data sets for classification and regression to approach a solution. ML, together with AI, has delivered a notable gain to the QoS requirements of URLLC by solving a problem in a distributed fashion, and also referred to as AI and ML on the edge and on the device.

5) *Grant free access*: In persistent scheduling, rapid access to the resources on high priority is mandatory. However, in semi-persistent unused resources must be allocated to the eMBB applications. In grant based access, the BS control the resources. However, in grant free access, the resource assignment procedure is skipped which eventually enhancing the latency of URLLC.

## VI. APPLICATIONS OF 5G-IIoT IN-TERMS OF URLLC TO eMBB

With the emergence of 5G/B5G, several IIoT applications such as smart cities, smart industry to smart grid and intelligent networks gain benefits from its distinguished features, e.g., URLLC and eMBB are two notable services for IIoT applications. eMBB fulfils the bandwidth requirement of data-intensive IIoT applications, e.g., AR, industrial video surveillance and VR. However, applications such as robotics, motion control, autonomous driving controls are time sensitive and mission critical applications and are, therefore, served by URLLC. Moreover, eMBB provides much higher data rate of up to Gbps with better coverage, while URLLC reduces the delay and provides the latency of 1 ms to the IIoT applications. Table IV provides IIoT applications along with their throughput, latency, and reliability bounds.

Various B5G-IIoT applications are shown in Fig. 8. These applications are self driving car, mission critical application, i.e., remote control robots and machines, time sensitive applications in which latency is a grave concern, industry automation AR, VR and HD-video. These applications are categorized from URLLC to eMBB. Each application has its own QoS requirement which needs to be satisfied via different enabling and communication technologies. These important IIoT applications are discussed herein.

### A. Smart Factory

Smart factory design parameters are mission-critical, with closed-loop control system which requires a latency of less than 1 ms and 99.99% service availability. Specifically, industrial and health robots with motion control are categorized as URLLC, as they are delay sensitive and requires high reliability and low latency. However, eMBB based AR and VR need high data rate and throughput. Therefore, in general B5G-IIoT provides communication between inter and intra industrial factories.

Certain use cases for smart factories are discussed in Table IV with corresponding parameters. These use cases include printing and packaging machines, remote control automation and monitoring, industrial robots, video and motion control, milling machines and cranes. Moreover, the latency and throughput requirements of the aforementioned use cases are also listed. URLLC use cases include automation, industrial robots, motion control [151]. These use cases seeks a latency of less than 10 ms and 99.9999% resources availability. While eMBB applications include virtual and augmented reality which needs high data rate universal connection [19], [152], [156], [163]–[167].

Smart Factory is an application of B5G-IIoT. To get maximum benefits from it, the key implementation and QoS maintaining issues are discussed. Smart factories have many interconnected networks, industries, and devices including industrial wireless sensor networks (WSNs) which needs connections with minimal delays, must be power efficient for continuous processing, and provide no queuing of data. To jointly exploit all these services and features in a smart factory is a big challenge for the research community. Licensed and unlicensed spectrum is used for communication between devices to gain maximum possible data rates. In B5G, the mm-Wave is considered a suitable choice for high throughput, reliability, and low latency. However, it suffers from various attenuation.

### B. Smart Autonomous Vehicles

Smart vehicles require an efficient real-time exchange of data acquisition and processing. Smart vehicles are designed with many sensitive sensors which can be efficiently controlled and monitored to fulfill the requirements of URLLC and eMBB. The efficient control of sensors ensures the safe exchange of real-time secure transfer of large volume of data at high rate. The latency requirement of less than 10 ms and a throughput of up to Mbps encourages to partially or fully shift the data processing to the cloud and the edge. To design a smart autonomous vehicle system, data sensing and transmission can be done from vehicle to vehicle and from vehicle to traffic using diverse communication technologies including ZigBee, WiFi, and 5G/B5G.

Several smart autonomous vehicle use cases are discussed in Table IV. These use cases are accidental warnings, high-speed trains, and automated driving. These use cases are time-sensitive and need a high-speed data exchange. For example, in an accidental scenario, ultra higher throughput with minimal latency is desired [154], [155], [168], [169].

### C. Smart Grid

Smart grid is among the most monitored and controlled IIoT applications. Owing to B5G, the smart grid generates, transmits, and distributes electricity with additional reliability, scalability, sustainability and safely. From consumer perspective, the usage of electricity is more adaptive in high speed vehicles, storage units, wireless sensors and IIoT devices. B5G-IIoT drives the smart grid more efficiently due to reduced delays and enhanced reliability. Moreover, B5G-IIoT connects

low-voltage grid stations to other elements of smart grid, and the backbone networks of B5G to medium and high power voltage grid stations.

Typical use cases for smart grid include medium and high electricity distribution stations, provided in Table IV. These use cases have different latency and throughput requirements according to their targets, i.e., future and sudden disturbance in normal grid station need less than 5 ms of latency and a data rate of up to 30 Mbps for up-link and 3 Mbps for down-link receptions. Thus, it is expected that up to 100 customers can be simultaneously supported with less than 1 ms delay and 99.999% resources availability. Similarly, B5G-IIoT uses mm-Wave for smart meters for accurate prediction of the energy usage of the consumers and to effectively transmit the information to the micro-grid [154], [155], [183]. The problems often occurring in the industrial smart grids are required to be timely controlled. Moreover, the seamless power transmissions between city areas need to be real-time, time-sensitive, and should be monitored and controlled to avoid disruptions in connectivity.

The Smart Vehicle IIoT applications suffer from wireless connectivity and other QoS related challenges. Some of the challenges like system performance reliability, connectivity, and passenger's internet availability are the main concerns. The process of acquisition and processing of real-time data in a small vehicle, global positioning system (GPS) connectivity, fuel, and hazard monitoring applications need efficient communication, while jitters and latency can cause serious hazardous situations. Thus, real-time data acquisition and time-sensitive applications are challenging to deal with. Allocation of resources in a smart vehicle is also challenging because of its mobility and constantly changing geographical locations. Thus, efficient implementation of V2X infrastructure with multi-RAT technology for low latency, high throughput, and high reliability makes it more challenging for smart vehicles.

#### D. Smart Cities

The concept of smart cities is trending from the last decade. Typically in a smart city, the services like, the transportation, health-care and schooling are integrated with the communication infrastructure. Traffic update is another modern time sensitive smart city application and, therefore, can be implemented using URLLC. Thus, IIoT plays a pivotal role in building smart cities by integrating different applications at high data rate, high reliability, and low latency. Communication technologies like LoRa wide area network (LoRaWAN), SigFox, WiFi, 802.11x, Bluetooth and LTE are used data collection and processing of data from smart city sensors'.

Use cases for smart city include managing of larger events like football matches, shopping malls and media. High-quality live coverage is needed for media activation and at ultra higher data rate. Similarly, health and traffic systems seek real-time monitoring and should be communicated with less latency and high reliability to the consumer end. B5G-IIoT supports high reliability, higher data rate with less than 1 ms latency for smart cities [155], [162], [168]–[173].

Smart Cities has many applications like smart homes, transportation, street lighting. Almost all these applications the QoS

requirements of URLLC and eMBB need to be implemented in different ways, using different communication and enabling technologies. Real-Time and time-sensitive monitoring and control applications, including accidental cases, highways, and traffic, are susceptible to low latency and high reliability. However, for large gatherings in shopping malls and stadiums, the real-time coverage with high throughput and low latency is required. Thus, joint optimization of multiple QoS constraints is a challenging task.

#### E. Smart Healthcare Service

New opportunities in healthcare improvement are discovered through the IIoT [184], [185]. Owing to ubiquitous identification, sensing, and communication capabilities in IIoT, all objects in the healthcare system can be continuously tracked and monitored [186]. These objects comprise the people, medical equipment, and medicine. Healthcare-related information can be collected, managed, and shared through global connectivity. This information is related to the logistics, diagnosis, therapy, recovery, medication, management, finance, and even the daily activities of the patients. For example, the doctors can periodically monitor the heart rate of a patient under an IIoT aided healthcare facility. The IoT-based healthcare services can be formed as mobile and personalized using personal computing devices, e.g., laptops, mobile phones, tablets, and mobile internet access [187]. The universal availability of mobile internet has accelerated the development of IoT-based at-home healthcare (AAH) services [185]. Security and privacy are the major concerns and challenges in smart healthcare services.

In smart hospitals, B5G-IIoT plays an important role and provides services for living and the health care system. Remote health monitoring of the devices and patients must be quick with minimum latency and high reliability. Similarly, a robot that assists in surgery requires minimum end-to-end latency, high reliability, and throughput which makes it more sensitive and challenging.

#### F. Smart Homes

Smart homes are intelligent buildings, equipped with smart devices and sensors that are connected to the internet, are able to make instantaneous and distributed decisions. Besides, the users can control the electrical appliances with smart and intelligent technologies, including bilateral computer interfaces (BCI) and AI. These technologies analyze human behaviors to create a safer, convenient, and easier environment for the end-users. The smart home will become a reality with 5G and B5G.

#### G. Smart Robots

Autonomous robotic systems allow the development of the capabilities of individual robots, their integration into platforms, and enable the development of fleets of heterogeneous robots for collaborative applications. Usually, robotic applications require robust and resilient wireless/cellular communication. The applications use a variety of communication

channels, including cellular, optical, audio, and video. In communication networks, protocols for many purposes may be used, including 802.15.1, 802.15.4, 802.11, 4G/LTE/5G, and more. Edge-distributed processing, instead of centralized processing, is more suited for mission- and safety-critical robotic applications. These applications include analytic based on AI methods. Edge-distributed processing makes use of multi-access edge and fog computing technologies, as well as wireless and cellular communication, i.e., 4G, 5G and B5G. The use of intelligent networks can facilitate the transfer and processing of information in an energy-efficient manner. The artificial neural networks (ANNs), ML, and other AI techniques further improve the efficiency of decentralized data analytic, automated network management, and context and knowledge sharing for robots and IIoT applications.

#### *H. Smart Industrial Manufacturing*

Industries are always in dire need of smart, efficient, and cost-effective manufacturing solutions to increase their revenues, and simultaneously to fulfill the market's demand and supply needs. Modern technologies including robotics, warehouse automation, and smart factories play a vital role in the industries to accomplish the manufacturing goals. These advancements are further enhanced through B5G wireless networks and IIoT. With B5G, manufacturing industries will be quickly transformed into smart industries by automation, AI, and AR techniques. The B5G revolution in industry digitization is upgrading connectivity, speed, quality, and latency. The B5G can successfully handle the manufacturing challenges, including bandwidth, speed, and latency. Owing to the benefits offered by B5G, factory floor machines and robots will be connected for better predictive maintenance. For factory floor layout changes and adjustments, B5G networks will offer higher flexibility, lower costs, and shorter lead times. With advanced B5G networks and information processing technologies, smart factories can be streamlined, internal and external communications can be improved, and product life cycle management can be unified on a single network.

#### *I. Smart Forestry*

Digital forestry is turned into smart forestry with characteristics such as perception, materialization, intelligence, and ecology. Research, production, management, and services in forestry are heavily dependent on private wireless communication networks. In the forest environment, wireless connectivity is the main challenge. Typically, distinct wireless communication technologies are used to overcome the challenges in smart forests. These technologies comprise GSM, general packet radio service (GPRS), Zigbee, bluetooth low energy (BLE), IPv6 over low-power wireless personal area networks (6LoWPAN), RFID, LoRa, Sigfox, NB-IoT, LTE, WiFi, NFC, and Z-wave. With the smart forestry, management and service values of forestry will be deeply explored and the forestry economy will become more prosperous.

#### *J. Smart Traffic Control*

Smart traffic management is susceptible to low latency and high reliability for proper management and observation of

traffic, between cities and on highways. Data exchange must be with high reliability and low latency to reduce hazards and emergencies. URLLC and eMBB plays an important role in road safety management [19], [153]–[155].

### VII. FUTURE RESEARCH DIRECTIONS

Despite the promises IIoT offers, there are challenges that needs to be addressed to utilize the full benefits. To fulfill the requirements of industry, some of the research gaps are identified and discussed in this section. The major future research topic in 5G/6G-IIoT are presented in [26], [44], [49], [74], [188]. Following are some of the research directions for IIoT that needs attention

#### *A. Dense IIoT Applications*

In B5G-IIoT, small applications are interconnected to make IIoT network. Research studies are required to make a dense IIoT network with feasible network infrastructure and integration of communication technologies in B5G-IIoT. B5G-IIoT refers to a network of connected objects that form a network of interconnected smart factories. Hardware refers to the physical existence of devices, while software refers to the capability of IIoT applications to provide services for everyone at the same time at different locations. It is not common for industrial devices, such as sensors, actuators, and controllers, to run learning algorithms. Nevertheless, smart industrial devices can potentially provide learning platforms by leveraging their computing capabilities. Learners can run learning tasks on those platforms that require fewer resources (e.g., computing, energy, etc.). By enabling the learning on industrial devices, the intelligence and performance of IIoT can be improved. A practical approach is to design a IIoT device in-loop with simulation platform for the internet of everything, which integrates computer simulations with IIoT device test beds. Simulating IIoT hardware testbeds from real-world data is what makes IIoT device in-loop simulation feasible. A further avenue is to develop an integrated simulation framework for IIoT that can capture the interactions and reciprocal effects between cyber and physical systems.

#### *B. Integration of Cloud and Edge Computing*

In B5G-IIoT, computing is a critical factor that affects latency, reliability, and other performance metrics. In order to provide storage, transmission, and analysis of industrial data, IIoT requires powerful and efficient computing platforms. In cloud computing, the computing resources are leveraged from the cloud to assist an industrial unit, on the other hand edge computing leverage computing resources from network edge devices. The hybrid cloud and edge computing platform provides IIoT computing infrastructure with the required capability and efficiency. Due to limitations on complexity, storage, and processing power, it may be impossible to carry out training on a device for a sufficiently complex learning problem. By reducing network latency through edge computing, it will be possible to cut the effectiveness of the learning process. Integrating edge computing infrastructures, especially



in an industrial context, improves resource utilization. Edge computing, single and multiple edge and cloud connectivity and computing boost the B5G-IIoT applications' performance both in terms of latency and reliability. The data analytic, storage, caching, and computing are yet the open research directions in B5G-IIoT. Open research problems also include implementing and optimizing self-organization, efficiency, and run-time features of cloud-based learning. The research also focuses on machine learning algorithms with configurable parameters and platforms that will aid decision-making in B5G-IIoT.

### C. Industrial Automation

Manufacturing, supply chain management, energy management, and human resources are all industrial processes that heavily rely on OT and their expertise. Combining AI and B5G-IIoT will allow these operational processes to be enhanced and taken to a whole new level of precision. The integration of B5G-IIoT with AI, big spectrum data and data analytic enhance the services of B5G-IIoT. AI in B5G-IIoT connect the automotive industry, smart homes and machine tools to make things digital and more productive.

Everything in today's world is data-driven, whether industrial processes or smart home devices. Industrialists are currently unable to manage the massive amounts of data generated in an industrial complex or the B5G-IIoT ecosystem as a whole due to a lack of skilled human resources and reliable tools for productively utilizing big industrial data. AI is capable of autonomously managing itself and will be able to overcome the limitations of skilled human resources or tools, allowing for optimal utilization and optimization. A connected IIoT ecosystem of devices coupled with AI-powered analytical models can enhance not only the manufacturing operation, but the entire industrial process as well. The combination of AI and IIoT presents too many advantages to ignore or overlook in terms of dependability and reliability.

Automated mobile robotss (AMRs) are becoming more important for manufacturing companies to support their logistics operations. An AMR can move flexibly in a factory and can recognize and avoid obstacles at the highest safety level, and they can be programmed easily. Furthermore, the following two factors need to be addressed as efficiently as possible: (a) unpredictable delivery times and downtime, and (b) vulnerability to network attacks. The fierce competition in IIoT and market maturity give many legacy companies in infrastructure management a stronghold throughout the value chain by selling software. Take advantage of a hybrid cloud designed to offer manufacturers and plant operators a more open, flexible, and secure alternative for capturing and analyzing real-time data from IIoT [188]. The industrial sector must pass this transformational and evolutionary phase in order to survive, and only those who gain the most from the change will be able to survive.

### D. Spectral and Energy Efficient Designs

Many stringent B5G-IIoT applications consume excessive energy and require high speed transmission as well. Thus,

spectrum sharing and making energy efficient B5G-IIoT applications give new directions to the research community. Numerous methods to study the spectral and energy efficient designs have been done for WSN and some for IoT network, however a collective effort for the design of IIoT network is still an open problem.

An interesting research direction is the use of unmanned aerial vehicle (UAV) which will replace a large number of IIoT devices, particularly in agriculture, traffic, and monitoring, which will reduce power consumption and pollution. In the future, UAVs are expected to provide low-cost, high-efficiency IIoT systems. Another more useful and a very hot and promising technology for B5G-IIoT is transmission of sensor data to the mobile cloud. Green social networking (GSN) as a service can investigate the system, service, wireless network, and cloud management in terms of energy efficiency. Another direction is the use of M2M communication to reduce energy usage and hazardous emissions in automated IIoT systems. We need appropriate techniques for improving QoS parameters (bandwidth, delay, and throughput) in order to enhance the SE of B5G-IIoT system. Another interesting research direction is to develop a green IIoT, which will utilize less energy, search for new resources, and minimize the impact of the IoT on human health and the environment. Green IIoT poses an important contribution to achieving a smart, sustainable, and green environment. RF energy harvest is of critical importance for an energy-balanced communication in B5G-IIoT. It is necessary to conduct more research in order to design IIoT devices that can reduce CO<sub>2</sub> emissions and energy consumption. Saving energy and reducing carbon dioxide emissions are critical to a green and sustainable life.

### E. Security and Privacy

Security and privacy solutions for B5G-IIoT is always an open research topic and it is deemed as one of the major obstacle in the wide spread adoptability of this network. Owing to the increase in the inter-connectivity of the IIoT network, there is a need for a robust and secure transmission between the nodes. Standard IoT solutions can in general be applied to some of the applications of IIoT, however, some sophisticated techniques are required for applications with stringent safety concerns and resource constraint. Research in the direction of network security, data sharing and security, authentication and security monitoring for specific applications of IIoT is required.

In order to support industrial systems, which require real-time information, e.g., sensor data, and commands, security mechanisms should guarantee both security services and timely communication. Since most B5G-IIoT systems rely on large amounts of data, storing data in a third-party system is logical. Data confidentiality and the privacy of important assets should be the top concerns regarding the transfer of data to a third party. Several potential solutions to these problems have been identified, including homomorphic encryption and searchable encryption, but more research is needed with regard to scalability, the delivery of data from the cloud in time-critical applications, and reliable recovery mechanisms

when third-party storage systems are compromised. Another important area for cryptographic research is the development of quantum-safe crypto-systems that scale with resource-constrained devices. A quantum-safe security scheme suitable for the IIoT is urgently needed, given the significance of the IIoT, and the benefits of public-key cryptography for managing key material for a large number of devices.

B5G-IIoT deployments often employ heterogeneous hardware and software solutions. This poses another research challenge. It is challenging to ensure that required security services are maintained as data spreads across multiple layers and an array of hardware and software. It is especially essential in situations where end-to-end encryption is required since data may travel across multiple hardware devices and software implementations, as well as across multiple vendors' hardware and software. Research into designing tamper-resistant hardware and maintaining operational safety in the face of such physical attacks are crucial for IIoT deployments in which an attacker may physically tamper with a device. It may be necessary to perform more research on the development of specific safety guarantees to be implemented with a complex IIoT system in safety-critical applications.

Another area of research is developing a privacy-preserving data collection and analytics approach. This can be accomplished by anonymizing data. Additionally, it may become necessary to verify the identities of devices by requesting that they self-identify without disclosing anything. Therefore, it is necessary to research and develop techniques for (pseudo) anonymization and zero-knowledge proofs that are suitable for the IIoT. The integration of the IIoT with other emerging technologies such as B5G-IIoT and blockchain is another important research direction.

#### *F. Machine/Deep Learning Solutions*

Many applications need real-time monitoring with closed-loop controls in which the industry can monitor itself and adapt automatically to provide benefits. This gives rise to a new research domain of the integration of industry with machine learning solutions. Many stringent B5G-IIoT applications consume excessive energy. Thus, spectrum sharing and making energy-efficient B5G-IIoT applications give new directions to the research community.

In B5G-IIoT, source space and target space have the same properties. However, complex IIoT systems in reality will not only undergo a domain shift, but will also possess different characteristics. A heterogeneous domain consists of similar IIoT systems manufactured by different manufacturers and equipped with different sensors. In the field of heterogeneous unsupervised domain adaptation, especially for complex physical systems, there is little research, but it has considerable potential, especially for IIoT applications. A second approach is to use simulation environments and then adapt them to real-life applications. The data in the source domain will most likely be sufficient for this direction. An open research question is how to evaluate the plausibility of the generated samples in terms of physical system coherence. Furthermore, controlling data generation from a physical process perspective also is an open research question.

In IIoT applications with extensive and detailed simulation environments, reinforcement learning (RL) has been especially successful. In addition, the traditional applications of RL have been comparatively simple. Further research is needed to extend to more complex problems. At present, there is no consensus or consolidation on how physics-induced ML can be applied to industrial applications, despite several directions being pursued. This research needs to be developed and consolidated to improve the interpretability of the developed models and methods. Most studies, however, focused on vibration data that was preprocessed into image-like signals. An important question is if these approaches can be extended to complex data sets and time series. The plausibility of the generated samples and the effect of the generated faults on the accuracy of the algorithms needs to be evaluated further. In future studies, the composition and selection of the training datasets must also be addressed. It is necessary to decide continuously if new measured data should be incorporated into the training data and algorithms updated, or if the data is redundant or already contained in the dataset used for training algorithms. In addition, this becomes even more complex if the decision to include the new measurement data is made on a fleet level in IIoT systems rather than solely on an individual unit level. Furthermore, the algorithm is improved by selecting those observations that will lead to the closest improvement in performance.

## VIII. CONCLUSION

IIoT has the potential to transform many aspects of our daily lives, including the environment, business, infrastructure, and lifestyle. This survey provides an in-depth assessment of IIoT, including the architecture and goals of the B5G-IIoT. The majority of the earlier literature focused on the architecture of IIoT. This primary focus of this paper is on the URLLC and eMBB techniques for 5G/B5G-IIoT applications. The trade-off between URLLC and eMBB for 5G-IIoT is presented, and insights for various applications are discussed. Future wireless communication is expected to play a significant role in optimizing the reliability and throughput of IIoT.

## APPENDIX I: ACRONYMS

|         |                                                      |
|---------|------------------------------------------------------|
| 2G      | second generation.                                   |
| 3G      | third generation.                                    |
| 3GPP    | third generation partnership project.                |
| 4G      | forth generation.                                    |
| 5G      | fifth generation.                                    |
| 6G      | sixth generation.                                    |
| 6LoWPAN | IPv6 over low-power wireless personal area networks. |
| AAH     | at-home healthcare.                                  |
| AI      | artificial intelligence.                             |
| AMR     | automated mobile robots.                             |
| ANN     | artificial neural network.                           |

|          |                                                     |           |                                                   |
|----------|-----------------------------------------------------|-----------|---------------------------------------------------|
| AP       | access point.                                       | IIoT      | industrial Internet of things.                    |
| AR       | augmented reality.                                  | IoT       | Internet of things.                               |
| AWGN     | additive white gaussian noise.                      | IP        | internet protocol.                                |
| B5G      | beyond 5G.                                          | IPv6      | internet protocol version 6.                      |
| BCI      | bilateral computer interfaces.                      | IT        | information technology.                           |
| BDA      | big data analytic.                                  | ITU       | International Telecommunication Union.            |
| BLE      | bluetooth low energy.                               |           |                                                   |
| BS       | base station.                                       | LoRa      | low range.                                        |
| CA       | carrier aggregation.                                | LoRaWAN   | LoRa wide area network.                           |
| CAN      | controller area network.                            | LPD       | low power devices.                                |
| CAPEX    | capital expenditure.                                | LPWA      | low power wide area.                              |
| CoMP     | coordinated multi-point transmission and reception. | LPWAN     | low power wide area network.                      |
| CPMS     | cyber-physical manufacturing systems.               | LTE       | long term evolution.                              |
| CPS      | cyber-physical systems.                             | LTE-M     | long term evolution for machine.                  |
| CSI      | channel state information.                          | M2M       | machine to machine.                               |
| D2D      | device to device.                                   | MAC       | medium access control.                            |
| DC       | dual connectivity.                                  | MES       | manufacturing execution system.                   |
| DL       | downlink.                                           | MIMO      | multiple-input multiple-output.                   |
| DMRS     | demodulation reference signals.                     | mIoT      | massive IoT.                                      |
| DNS      | domain name system.                                 | ML        | machine learning.                                 |
| DSS      | dynamic spectrum sharing.                           | mm-Wave   | millimeter wave.                                  |
| EASDF    | edge application server discovery function.         | mMTC      | massive MTC.                                      |
| EE       | energy efficiency.                                  | MTC       | machine type communication.                       |
| eMBB     | enhanced mobile broadband.                          | Multi-TRP | multiple transmission and reception architecture. |
| EPS      | evolved packet system.                              | NAN       | neighbourhood area network.                       |
| EtherCAT | ethernet for control automation technology.         | NB-IoT    | narrowband Internet of things.                    |
| FD       | flexible data-rate.                                 | NFC       | near field communication.                         |
| FDD      | frequency division duplex.                          | NPN       | non-public networks.                              |
| gNB      | next generation nodeB.                              | NR        | new radio.                                        |
| GNSS     | global navigation satellite system.                 | NSA       | non-standalone.                                   |
| GPRS     | general packet radio service.                       | OEM       | original equipment manufacturer.                  |
| GPS      | global positioning system.                          | OPEX      | operating expenditures.                           |
| GSM      | global system for mobile communications.            | OT        | operational technology.                           |
| GSN      | green social networking.                            | P2P       | peer to peer.                                     |
| HARQ     | hybrid automatic repeat request.                    | PDCCH     | physical downlink control channel.                |
| HART     | highway addressable remote transducer.              | PDSCH     | physical downlink shared channel.                 |
| HD       | high definition.                                    | PDU       | protocol data unit.                               |
| IAB      | integrated access and backhaul.                     | PHY       | physical layer.                                   |
|          |                                                     | PoS       | point of sale.                                    |
|          |                                                     | ProfiBus  | process field bus.                                |
|          |                                                     | ProfiNet  | process field net.                                |
|          |                                                     | PSM       | power saving mode.                                |
|          |                                                     | PUCCH     | physical uplink control channel.                  |
|          |                                                     | PUSCH     | physical uplink shared channel.                   |
|          |                                                     | QoS       | quality of service.                               |

|        |                                           |
|--------|-------------------------------------------|
| RAN    | radio access network.                     |
| RAT    | radio access technology.                  |
| RF     | radio frequency.                          |
| RFID   | radio-frequency identification.           |
| RL     | reinforcement learning.                   |
| RRC    | radio resource control.                   |
| RTT    | round trip time.                          |
| SA     | standalone.                               |
| SCADA  | supervisory control and data acquisition. |
| SDN    | software-defined-networking.              |
| SE     | spectral efficiency.                      |
| SERCOS | serial real-time communication system.    |
| SNR    | signal to noise ratio.                    |
| TCP    | transmission control protocol.            |
| TDD    | time division duplex.                     |
| TSC    | time-sensitive communications.            |
| TSN    | time-sensitive networking.                |
| TTI    | transmission time interval.               |
| UAV    | unmanned aerial vehicle.                  |
| UDN    | ultra dense small cell network.           |
| UE     | user equipment.                           |
| UL     | uplink.                                   |
| UNB    | ultra narrow band.                        |
| URLLC  | ultra-reliable low latency communication. |
| URSP   | user-equipment route-selection policy.    |
| V2I    | vehicle-to-infrastructure.                |
| V2N    | vehicle-to-network.                       |
| V2V    | vehicle-to-vehicle.                       |
| V2X    | vehicle-to-everything.                    |
| VR     | virtual reality.                          |
| Wi-SUN | wireless smart ubiquitous network.        |
| WiFi   | wireless fidelity.                        |
| WLAN   | wireless local area network.              |
| WNAN   | wireless neighbourhood area network.      |
| WPAN   | wireless personal area network.           |
| WSN    | wireless sensor network.                  |
| WWAN   | wireless wide area network.               |
| ZigBEE | zonal intercommunication global-standard. |

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